

Ishrat Jahan Eliza

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RESEARCH INTEREST

Data Visualization - Empirical AI & Large Language Models in Visualization - Human-Centered Computing

EDUCATION

Graduate Research Assistant (PhD)

Kahlert School of Computing & SCI Institute, University of Utah

Salt Lake City, UT 84112

August 2023 - Current

- **Supervisor:** [Dr. Valerio Pascucci](#)
- **Course Highlights:** HCI, Visualization for Data Science, AI, Graduate Algorithm, ML

MSc in Computer Science

The University of Utah

201 Presidents Cir, Salt Lake City, UT

84112, USA

August 2023 - December 2025

- **CGPA:** 3.98/4.00

BSc in Computer Science and Engineering

Bangladesh University of Engineering and Technology

Dhaka-1000, Bangladesh

Feb 2017 - May 2022

- **CGPA:** 3.51/4.00
- **Final two-year CGPA:** 3.816
- **Thesis:** [eDakterBari - A Human-Centered Solution Enabling Online Medical Consultation for Resource-Constrained Communities in Bangladesh](#)
Supervisor: [Dr. A. B. M. Alim Al Islam](#)

RESEARCH EXPERIENCE

Publications

Animating Petascale Time-varying Data on Commodity Hardware with LLM-assisted Scripting

LLM and Large Scale Data

2026 the 11th International Conference

on Big Data Analytics (ICBDA)

- **Authors:** [Ishrat Jahan Eliza](#), Xuan Huang, Aashish Panta, Alper Sahistan, Zhimin Li, Amy Gooch, Valerio Pascucci
- **DOI:** [10.48550/arXiv.2603.07053](#)
- **Supervised by:** [Dr. Valerio Pascucci](#)
- **My contribution:** I designed an agentic pipeline with an LLM using the OpenAI API that can understand natural-language user queries and return the necessary parameters to create a time-series animation from real-world large-scale data.

Accessible Text Description for UpSet Plots

Visualization and HCI

EuroVIS'25

- **Authors:** Andrew McNutt[†], Maggie K McCracken[†], [Ishrat Jahan Eliza](#)[†], Daniel Hajas, Jake Wagoner, Nate Lanza, Jack Wilburn, Sarah Creem-Regehr, Alexander Lex
- [†]The first three authors contributed equally
- **DOI:** [10.48550/arXiv.2503.17517](#)
- **Supervised by:** [Dr. Alexander Lex](#)
- **My contribution:** I conducted and designed a qualitative analysis of the UpSet plot from academic publications to generate a template of human-readable text descriptions in collaboration with our blind co-author. I used Python to generate the text description for the UpSet plot. The Python package is published on [PyPi](#).

eDakterBari: A Human-Centered Solution Enabling Online Medical Consultation and Information Dissemination for Resource-Constrained Communities in Bangladesh

Journal: Heliyon

Human-Centered Computing

- **Authors:** Ishrat Jahan Eliza, Mobasshira Akter Urmi M.D. Tousif Tanjim Anan, Tanveer Hossain Munim, Fattah-Zul-Ikram Galib, A.B.M. Alim Al Islam
- **DOI:** [10.1016/j.heliyon.2023.e23100](https://doi.org/10.1016/j.heliyon.2023.e23100)
- **Volume:** 10, **Issue:** 1, Elsevier (2024)
- **Supervised by:** Dr. A. B. M. Alim Al Islam
- **My contribution:** eDakterBari is a platform for providing health services to orphans, a community in Bangladesh with limited resources. I developed a mobile application in Flutter and Dart to support healthcare consultation for orphans in Bangladesh via intermediation. Conducted semi-structured surveys and analyzed data using Content Analysis.

A bimodal longitudinal investigation on changes in sentiments over social media interactions owing to COVID-19 pandemic

Journal: Scientific Reports

Natural Language Processing and Social Media Analysis

- **Authors:** Md Saidul Hoque Anik, Nuwaisir Rabi, **Ishrat Jahan Eliza**, Md Hasibul Husain Hisham, Ajwad Akil, Abir Mohammad Turza, Farhan Feroz, Fahim Morshed, Nazmus Sakib, Jannatun Noor, Sriram Chellappan, A.B.M. Alim Al Islam
- **DOI:** [10.1038/s41598-025-44070-0](https://doi.org/10.1038/s41598-025-44070-0)
- **Volume:** 15, **Issue:** 1, **Pages:** 44070, Nature Publishing Group UK (2025)
- **Supervised by:** Dr. A. B. M. Alim Al Islam and Dr. Sriram Chellappan

Revealing Influences of Socioeconomic Factors over Disease Outbreaks

COMPASS '22: ACM SIGCAS/SIGCHI

Exploratory Data Analysis

- **Authors:** S Mahmudul Hasan, Alabi Mehzabin Anisha, Rudaiba Adnin, **Ishrat Jahan Eliza**, Ishika Tarin, Sadia Afroz, A. B. M. Alim Al Islam
- **DOI:** [10.1145/3530190.3534804](https://doi.org/10.1145/3530190.3534804)
- **Supervised by:** Dr. A. B. M. Alim Al Islam

Note: CORONOSIS: Corona Prognosis via a Global Lens to Enable Efficient Policy-making Both at Global and Local Levels

COMPASS '22: ACM SIGCAS/SIGCHI

Mathematical Computing and Statistical Analysis

- **Authors:** **Ishrat Jahan Eliza**, Md Hasibul Husain Hisham, Mohammad Nuwaisir Rahman, Ajwad Akil, Abir Mohammad Turza, Fahim Morshed, Nazmus Sakib, Sriram Chellappan, A. B. M. Alim Al Islam
- **DOI:** [10.1145/3530190.3534831](https://doi.org/10.1145/3530190.3534831)
- **Supervised by:** Dr. A. B. M. Alim Al Islam and Dr. Sriram Chellappan
- **My contribution:** I conducted statistical analysis in Python to identify relationships between various environmental, socioeconomic, and COVID-19 variables (death, affected). In collaboration with the other members of the data analysis team, I performed prediction analysis using the epidemiological model SIRD. Later, to make it accessible online, I deployed in AWS.

ACADEMIC SERVICE

2025 **Reviewer**, IEEE Visualization and Visual Analytics Conference

IEEE VIS

2026 **Reviewer**, 29th Eurographics Conference on Visualization

EuroVIS

RESEARCH PROJECTS

Agents for Scientific Dataset Exploration

LLM and Visualization | Manuscript in Preparation

- Designing a framework using a Large Language Model for exploring scientific datasets through natural language with visualization generation, and textual insights.

Evaluating Visualization Modalities for Tumor Margin Localization

Medical Visualization and HCI | Ongoing

- Conducting a comparative user study evaluating different display modalities on surgeons' ability to localize tumor margins in 3D tissue models to inform surgical visualization tool design.

User Interaction Analytics Framework

Human-Centered Computing | Ongoing

- Developing a user analytics framework that captures and replays user interactions on whole slide images, with a dashboard for behavioral analysis to support inference training models

User Study to Evaluate Text Descriptions for UpSet Plots

Visualization and HCI | Published

- Running a within-subject survey and semi-structured interviews via [ReVISit](#) on Prolific, targeting researchers across the USA, UK, and Canada. Analyzing quantitative and qualitative data using SciPy, Vega-Altair, and thematic analysis.

WORK EXPERIENCE

IQVIA Asia Pacific

Dhaka, Bangladesh

Software Developer

June 2022 - July, 2023

- Worked with the Data Science team in collaboration to develop and investigate various potential algorithms that might be used to report the market potential of the medical products of organizations around the world. I was part of the Python team.
- Tested the existing potential algorithms running in the backend service by implementing unit test cases in Python.
- Technical Skills:** Python, Git
- Soft Skills:** Communication, Time management, Logical Thinking.

Free Pixel Games Ltd.

Dhaka, Bangladesh

Data Analyst Intern

December 2020 - January 2021

- Analyzing game data, including user, session, game level, and other activity data.
- Technical Skills:** MySQL, Tableau
- Soft Skills:** Communication, Logical Thinking.

SKILLS

Programming	Python, JavaScript, C/C++, Java, Dart, HTML/CSS, R
Frameworks/Libraries	Pandas, NumPy, ReactJS, NodeJS, Flutter, Keras, MongoDB, MySQL, NextJS
Tech	Git, Postman, Latex, VS code, Excel, Adobe Premiere Pro, MS Office, Figma
Data Analysis/ Visualization	SciKit-Learn, Seaborn, Vega-Altair, Matplotlib, D3.JS, SciPy

EXTRA-CURRICULAR EXPERIENCE

2026	Lead Student Volunteer , The International Conference for High Performance Computing, Networking, Storage, and Analysis	SC'2026
2025	Student Volunteer , The International Conference for High Performance Computing, Networking, Storage, and Analysis	SC'2025
2024 - Present	Cultural Secretary , Bangladesh Student Association at the University of Utah	UofU
2020-21	President , BUET Dance Club	BUET
2017-22	Class Representative , Department of Computer Science and Engineering	BUET

REFERENCE(S)

Dr. Valerio Pascucci

Professor, SCI Institute, School of Computing
University of Utah ([UofU](#))
Email: pascucci.valerio@gmail.com
[\[Website\]](#), [\[Google Scholar\]](#)